

THORACIC ONCOLOGY — MULTIMODAL TEST CASE

SYNTHETIC TRAINING/QA DOCUMENT — ALL PATIENT DATA AND IMAGES ARE FICTIONAL

Patient ID	SYN-THX-2026-0417	Age/Sex	63 / Female
Accession	SYN-PATH-0810-417	Specimen	RUL core biopsy
Clinical Service	Thoracic Oncology	Document Type	Pathology + imaging case

1. Clinical Presentation

A 63-year-old woman presents with persistent cough. Chest CT demonstrates a 3.1 cm spiculated lesion in the right upper lobe with an adjacent small area of pleural retraction. Mild right hilar lymph-node enlargement is described. PET imaging reportedly shows increased FDG uptake within the dominant pulmonary lesion. A core biopsy is submitted for histopathologic evaluation.

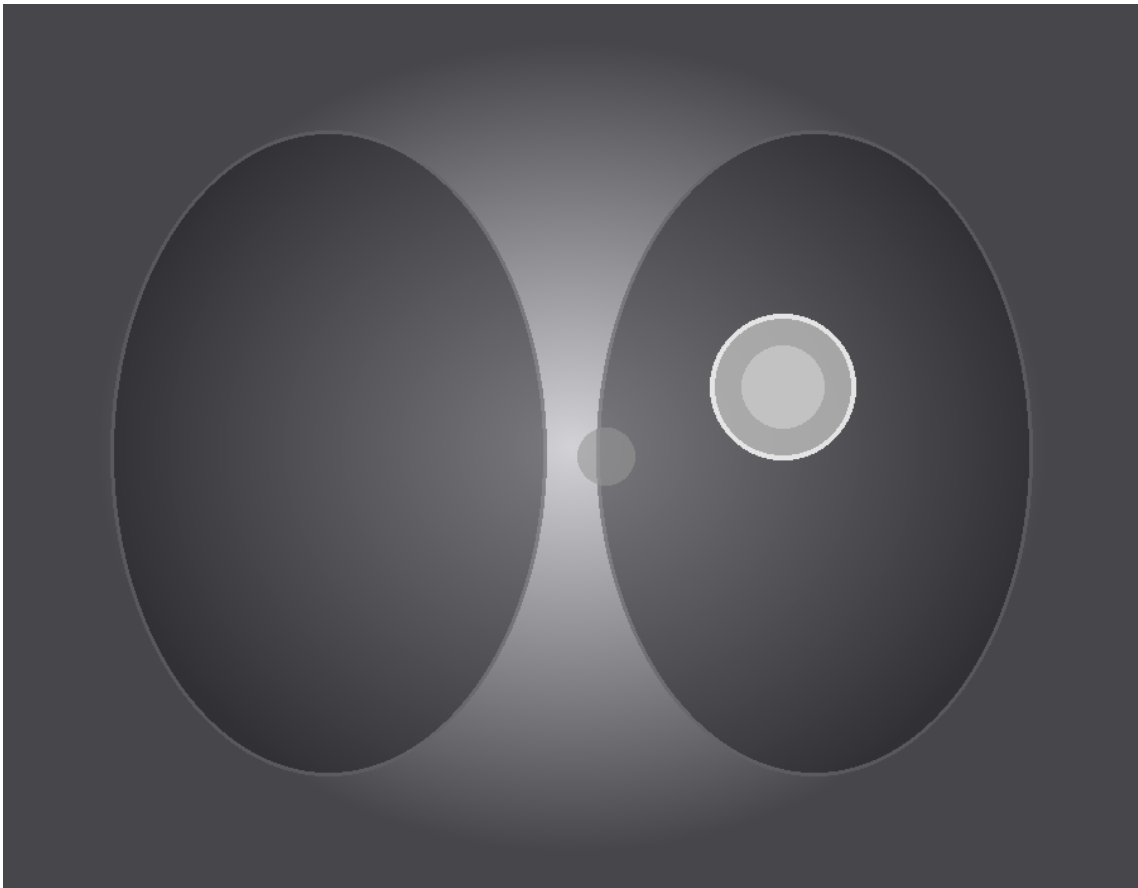


Figure 1. Simulated axial CT-style image created solely for AI pipeline testing. The circled-appearing high-density lesion represents the synthetic right-upper-lobe mass.

2. Specimen and Gross Description

Received in formalin are multiple tan-white, soft-tissue core fragments measuring 1.4 × 0.8 × 0.3 cm in aggregate. The specimen is submitted entirely in blocks A1–A2.

3. Microscopic Findings

Sections demonstrate an infiltrative malignant epithelial neoplasm. The tumor shows a mixture of gland-forming/acinar architecture and lepidic growth, with smaller foci of papillary and solid architecture. Tumor cells have enlarged ovoid nuclei, finely granular chromatin, occasional nucleoli, and moderate pale eosinophilic cytoplasm. Scattered cells contain intracytoplasmic mucin. Desmoplastic stromal reaction is present. Focal lymphovascular invasion is suspicious but cannot be confirmed with certainty in this limited biopsy.

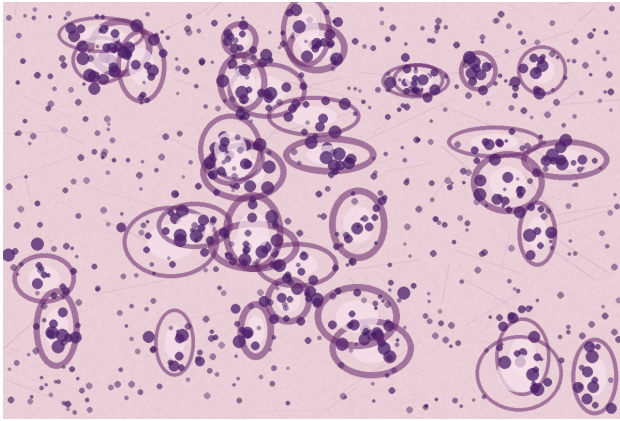


Figure 2A. Simulated H&E-style; field demonstrating an acinar/gland-forming pattern.

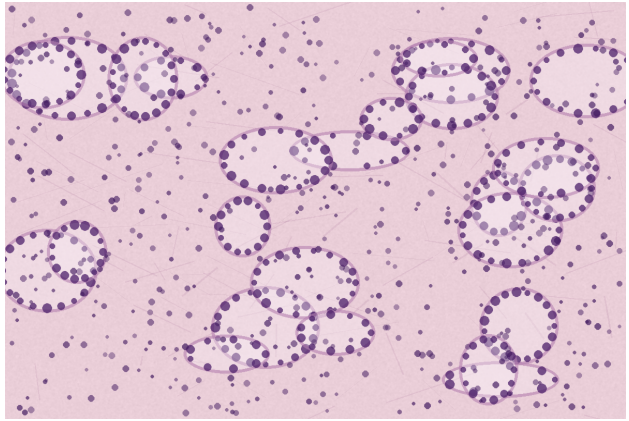


Figure 2B. Simulated H&E-style; field demonstrating lepidic-type growth.

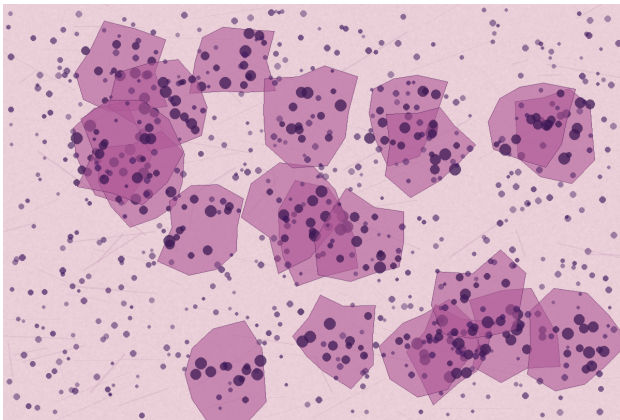


Figure 2C. Simulated H&E-style; field demonstrating solid architecture.

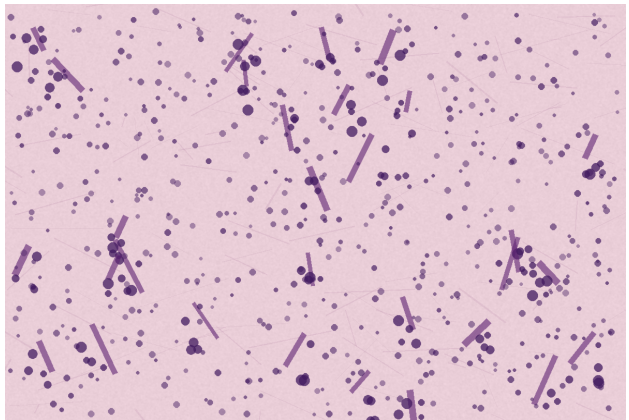


Figure 2D. Simulated mixed-pattern field with papillary/solid-type architecture.

4. Immunohistochemistry

Marker	Result	Interpretation
TTF-1	Positive	Nuclear staining in tumor cells
Napsin A	Positive	Cytoplasmic staining in tumor cells
CK7	Positive	Diffuse tumor-cell staining
p40	Negative	No convincing tumor-cell staining
CK5/6	Negative	No convincing tumor-cell staining
ALK (D5F3)	Negative	No expression detected by IHC screening

5. Final Pathologic Impression

RIGHT UPPER LOBE LUNG, CORE BIOPSY:

Non-small cell lung carcinoma, favor adenocarcinoma.

The sampled tumor demonstrates predominantly acinar and lepidic morphology with focal papillary and solid patterns. The immunophenotype supports pulmonary adenocarcinoma in the appropriate clinicoradiologic setting.

6. Ancillary / Molecular Testing

Test	Status
EGFR mutation analysis	Pending
KRAS mutation analysis	Pending
ROS1 rearrangement	Not performed
ALK	Negative by IHC screening
PD-L1 tumor proportion score	Pending

7. Case Limitations

This is a limited core biopsy. The relative proportions of histologic growth patterns cannot be reliably estimated for the entire tumor. The suspicious focus of lymphovascular invasion requires correlation with additional tissue if clinically relevant. Imaging

and clinical findings should be interpreted together with the pathology.

8. Questions for Oncology AI Evaluation

Q1. What is the most likely diagnosis based on the report and images?

Q2. Which histologic growth patterns are described or illustrated?

Q3. Which immunohistochemical findings support pulmonary adenocarcinoma?

Q4. Which findings are negative, and which molecular tests remain pending?

Q5. What limitations should be considered when interpreting this biopsy?

Q6. Does the report establish a complete tumor stage? Explain why or why not.

Q7. Compare the textual pathology findings with the illustrated histologic patterns. Identify any mismatch rather than assuming the images prove a diagnosis.

AI TESTING NOTICE: This entire report, including the microscopy-style and CT-style images, is synthetic and was generated for software QA/testing. The images are illustrative rather than real patient histology or radiology. They must not be used for clinical diagnosis, treatment, education as authentic pathology, or research involving human subjects.